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Review Article

Machine learning for food flavor prediction and regulation: models, data integration, and future perspectives

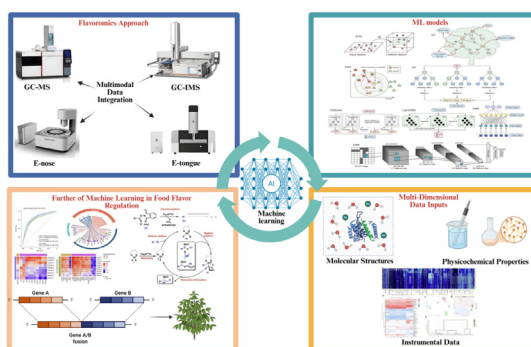
Xinyu Ge, Yongjie Zhou, Qing Li, Yuqing Tan, Yongkang Luo, Hui Hong*

Beijing Laboratory for Food Quality and Safety, College of Food Science and Nutritional Engineering, China Agricultural University, Beijing 100083, China

HIGHLIGHTS

- Introduces the advantages of flavoromics and multimodal data in Machine learning (ML)
- Introduces and compares commonly used ML for food flavor research.
- Outlines ML-based application to flavor prediction and regulation in food systems.
- Explores future prospects of ML in food flavor regulation.
- ML-driven strategies enable targeted metabolite discovery and flavor regulation.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: Flavor is a central attribute of food quality, shaping consumer preferences and market performance. Traditional evaluation methods, such as sensory panels and basic assays, are often constrained by subjectivity, low throughput, and limited scalability. With the rise of high-throughput technologies and multimodal datasets, machine learning (ML) has emerged as a promising tool for deciphering and regulating complex flavor systems.

Aim of review: This review examines current flavor detection techniques and the application of ML across diverse domains. It compares supervised learning models (SVM, DT), ensemble algorithms (XGBoost, LightGBM), and deep learning approaches (CNN, ANN). This review also discusses the contribution of three major data dimensions to flavor prediction, as well as future prospects in the field. ML enables precise flavor prediction, compound screening, and real-time process control. To support these tasks, researchers have developed integrated analytical systems that combine electronic nose (E-nose), electronic tongue (E-tongue), gas chromatography–mass spectrometry (GC–MS), and gas chromatography–ion mobility spectrometry (GC–IMS). Ensemble learning and deep learning models show strong performance when handling complex, nonlinear datasets. Explainable artificial intelligence (XAI) tools such as Shapley Additive Explanations (SHAP) improve model transparency by linking predictions to underlying features. ML models further enhance both prediction accuracy and generalizability. Innovations such as attention mechanisms, graph neural networks, and digital twins support dynamic flavor modulation. ML also aids in identifying key flavor compounds and genotype-phenotype relationships, accelerating breeding and formulation.

Key scientific concepts of review: ML is opening up new technological avenues in flavor science, with significant potential to predict and control flavor formation mechanisms, verify product authenticity, and support the targeted design of flavor-active compounds that align with consumer expectations for sensory appeal.

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* Corresponding author at: 17 E Qinghua Rd, Iaidian District, Beijing, China.

E-mail address: hhong@cau.edu.cn (H. Hong).

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Introduction

In recent years, rising living standards have increased public expectations for food quality. Flavor enhancement has consequently become a central objective in product development. Flavor, which encompasses both olfactory and gustatory perceptions, is widely recognized as a core indicator of food quality and consumer acceptance [1]. Volatile compounds, such as aldehydes, ketones, and alcohols, primarily contribute to aroma. In contrast, non-volatile components, including amino acids, peptides, nucleotides, and organic acids, are key determinants of taste development [2,3]. Studies show that flavor reflects intrinsic food attributes such as origin, variety, and maturity. It also plays a decisive role in shaping consumer preference and purchasing behavior [4]. A desirable flavor profile enhances appetite and purchase intent, while poor flavor may suppress consumption. Therefore, precise prediction and effective modulation of food flavor are critical for product success, calling for the support of advanced scientific and technological tools. With rapid advances in computational technologies, artificial intelligence (AI), particularly machine learning (ML) as a key subfield, has opened new opportunities for flavor prediction and regulation in the food industry [5].

Conventional sensory evaluation methods, such as trained expert panels and consumer tests, have long been considered the reference standard for flavor assessment because they directly capture human perception. However, these approaches are constrained by several inherent shortcomings. Outcomes are often

affected by assessor fatigue, adaptation, and inter-individual variability, which undermine reproducibility. Consumer-based tests, while valuable, are expensive and time-consuming, and results are difficult to standardize across different cultural and demographic groups. More importantly, these methods provide only descriptive sensory outcomes and rarely reveal the chemical or mechanistic basis of flavor perception. Such drawbacks limit their scalability and make them insufficient to support high-throughput product development or predictive studies [6]. The emergence of flavoromics represents a major breakthrough in flavor research by enabling in-depth molecular-level characterization of flavor-active compounds. Through approaches such as solvent extraction and headspace techniques, in combination with advanced analytical instruments, researchers can identify and screen the key compounds driving flavor perception [7]. Common analytical platforms include gas chromatography-mass spectrometry (GC-MS), gas chromatography-ion mobility spectrometry (GC-IMS), electronic nose (E-nose), and electronic tongue (E-tongue). GC-MS enables high-precision detection of volatile compounds, while GC-IMS offers fast response and operational simplicity. The e-nose and e-tongue simulate sensory functions and allow rapid evaluation of overall flavor profiles. Although they cannot resolve individual compound structures, they are valuable for flavor profiling and rapid quality assessment [8]. Studies have shown that multimodal data integration, by combining subjective sensory evaluation with objective instrumental analysis, provides a more comprehensive understanding of the link between chemical composition and sen-

Table 1

Applications of flavoromics in food flavor analysis.

Analytical Method	Food Type	Aim of the study and findings	Reference
GC-MS	Peach (3 cultivars)	Detected decline in esters and increase in alcohols/aldehydes during storage	Li et al. (2025)
	Commercial milk powder	Identified 86 volatile compounds	Zhou et al. (2025)
	Golden pompano (various cooking methods)	Detected 108 aroma compounds	Chen et al. (2024)
GC-IMS	Sun-dried citrus peel	Detected 100 + peaks; identified 82 VOCs incl. aldehydes, alcohols, terpenes	Wang et al. (2023)
	Fermented black beans	Detected 91 VOCs; 23 odorants highlighted via ROAV	Miao et al. (2024)
	Golden pompano	Detected 72 VOCs; supported GC-MS findings	Chen et al. (2024)
E-nose	Fermented goat and lamb meat	PCA clustered samples by meat type	Zhang et al. (2025)
	Golden pompano	E-nose classified samples by cooking method	Feng et al. (2022)
	Freshwater fish (4 species, raw and cooked)	To characterize and discriminate VOC profiles in raw and cooked fish using E-nose	Chen et al. (2023)
E-tongue	Raw milk	To identify regional flavor differences in raw milk using e-tongue	Chi et al. (2024)
	Black mulberry juice	To evaluate how fermentation type influences taste of mulberry juice via e-tongue and metabolomics	Lv et al. (2025)
GC-MS/GC-I-MS + E-nose/E-tongue	Zhenjiang vinegar	55 VOCs identified by GC-MS; 14 linked to grade; e-nose sensors correlated to key volatiles	Wang et al. (2021)
	Crataegi fructus (roasted hawthorn)	To analyze the changes in volatile compounds during roasting using E-nose and HS-GC-MS	Fei et al. (2022)
	Grilled lamb shashliks (five commercial brands)	To characterize flavor profiles of grilled lamb shashliks using E-nose, E-tongue, and multiple GC-based techniques combined with chemometric modeling for VOC prediction and brand discrimination	Shen et al. (2023)
	Yak jerky	To compare flavor profiles of yak jerky from three muscles using E-nose, E-tongue, GC-MS, and GC-IMS, and identify key differential aroma markers	Zhou et al. (2024)

sory perception than any single method alone [6]. Overall, while sensory evaluation remains indispensable, its reliance on human judgment makes it unsuitable for high-throughput or mechanism-oriented studies. These limitations justify the search for complementary data-driven methods. Approaches such as flavoromics and advanced instrumental analysis have partly addressed these issues, yet they remain descriptive rather than predictive.

With the rapid development of flavoromics, multi-omics integration, and high-throughput analytical technologies, food flavor research has fully entered a new data-driven era [9]. There is an urgent need to develop efficient, automated, and scalable frameworks for flavor analysis and control. ML has demonstrated considerable potential in this context as a powerful tool for handling complex, high-dimensional flavor data. The application of machine learning to flavor has progressed in distinct stages. In 1980, Kier first proposed a ligand-based classifier for taste, which laid the foundation for subsequent studies on quantitative structure–taste relationships [10]. Keller et al. (2017) advanced the field through the DREAM Olfaction Prediction Challenge, where large-scale models were trained to predict human olfactory percepts directly from molecular features. In 2019, the introduction of convolutional neural networks (CNNs) into odor prediction, exemplified by POP-CNN, enabled representation learning from structural and spectral inputs [11]. More recently, explainable AI has been incorporated into flavor modeling. For example, Ferri et al. (2024) employed SHAP for interpretable modeling of taste perception [12]. Together, these milestones illustrate the transition from classical QSAR approaches to deep and interpretable learning in flavor science.

ML models can integrate three key data dimensions in flavor science: molecular structures, physicochemical properties, and instrumental outputs. By combining these sources, they enable the construction of predictive frameworks that elucidate flavor formation mechanisms, pinpoint key flavor compounds, and support flavor quality optimization [7,13]. Moreover, ML supports a wide range of tasks, including classification, clustering, regression, and rule extraction, and can continually refine its algorithms to identify complex data patterns and achieve accurate predictions for new datasets [14,15]. Researchers can select suitable algorithms according to their specific study objectives to improve model performance and generalizability. Various machine learning algorithms have been applied to flavor modeling, including support vector machines (SVM), decision trees (DT), ensemble learning models XGBoost, LightGBM, and deep neural networks (DNNs) such as artificial neural networks (ANN) and convolutional neural networks (CNN). All have shown robust predictive performance [16,17]. For instance, Elamshity et al. (2024) employed an ANN model combined with near-infrared spectroscopy and pH data to achieve high-precision flavor prediction in dairy beverages [18]. Similarly, Schreurs et al. (2024) applied a gradient boosting machine (GBM) model to GC-MS and sensory data from 250 beer samples, which not only successfully predicted flavor scores but also identified a series of key volatile compounds [14]. In recent years, flavor research has gradually shifted from passive analysis to proactive modulation, and the scope of machine learning in flavor control continues to expand. Researchers have begun employing ensemble learning and deep neural network models to screen key flavor metabolites from complex metabolomic datasets and

to establish quantitative relationships between these compounds and sensory attributes. For example, ML-driven metabolite screening has identified γ -dodecalactone as a key aroma contributor in strawberries. Deep neural networks (DNN) and reinforcement learning (RL) have also been successfully applied to flavor-oriented breeding (prediction of desirable tomato genotypes) and intelligent processing control (dynamic regulation of coffee roasting) [19–21]. In addition, emerging technologies such as deep learning and digital twins are being gradually integrated into real-time flavor control systems in food processing, enabling closed-loop management from raw material characterization to process optimization and final product flavor presentation. These advances not only improve prediction accuracy and flavor consistency but also offer new strategies for personalized food development and next-generation innovation [22]. Nevertheless, despite promising progress, there remains a lack of systematic guidance on how to select appropriate ML models and apply them effectively in flavor prediction and regulation.

Therefore, this review systematically summarizes representative studies and key developments in the application of machine learning to food flavor prediction and modulation. It covers a wide range of algorithmic approaches, including traditional supervised learning methods, ensemble learning strategies, and deep learning frameworks, and evaluates their practical use in different food systems. Additionally, it discusses emerging trends in multimodal data fusion, model interpretability, and four prospective application areas. This work aims to provide both researchers and industry professionals with a comprehensive theoretical foundation and practical reference, supporting the transformation of food flavor research from empirical strategies to intelligent, predictive systems.

Multimodal approaches for flavor characterization

Flavoromics is a comprehensive strategy that combines detailed chemical profiling with advanced data analytics to understand the flavor characteristics of food. In practice, it integrates techniques like GC–MS and GC–IMS with artificial sensory tools such as E-noses and E-tongues. These tools, supported by multivariate statistical models, help bridge the gap between measurable chemical data and how flavors are perceived by humans (Table 1).

Chromatographic platforms for aroma analysis

Among analytical tools, GC–MS and GC–IMS are mainstays in the study of food volatiles. GC–MS, particularly when paired with headspace solid-phase microextraction (HS–SPME), is a gold standard in food flavor research. It can identify a wide range of aroma compounds—esters, alcohols, aldehydes, ketones—within complex food systems [23]. However, GC–MS often involves time-consuming sample prep. GC–IMS, on the other hand, is faster and doesn't require pretreatment. It couples a short GC column with an ion mobility detector to produce visually intuitive two-dimensional spectra [24]. In many studies, GC–MS is used to identify specific compounds, while GC–IMS provides rapid fingerprinting, and together these methods offer a more complete understanding of flavor complexity. In recent years, mass spectrometry and intelligent sensory instruments have played an increasingly important role in food flavor analysis [25]. These tools allow for the efficient extraction, detection, and integration of both volatile and non-volatile compounds in food, providing essential data that reflects flavor-related characteristics.

Gas chromatography-mass spectrometry (GC–MS)

GC–MS is a hyphenated analytical technique that first separates volatile compounds in a complex food matrix by gas chromatogra-

phy and then identifies them based on their mass-to-charge ratio using mass spectrometry. After chromatographic separation, analytes are typically ionized by electron impact, fragmented, and subsequently detected in the mass analyzer to generate characteristic spectra that allow both qualitative identification and quantitative determination of flavor compounds.

When coupled with techniques such as headspace analysis or solid phase microextraction, GC–MS provides both high sensitivity and detailed molecular information [26].

For instance, Li et al. (2025) used GC–MS to study three peach cultivars during storage at 25 °C and 4 °C. The data revealed a sharp drop in total volatiles over time, particularly a decline in esters (responsible for fruity notes) under refrigeration, while alcohols and aldehydes increased. Using orthogonal Partial least squares discriminant analysis (PLS-DA), the researchers identified a mid-storage phase as a critical turning point in aroma quality, where ester degradation and the rise of compounds like hexanol and nonanal signaled flavor shifts among different cultivars [27]. In dairy products, Zhou et al. (2025) profiled 86 volatile compounds in milk powder, using multivariate analysis to cluster products by brand and highlight key markers such as medium-chain aldehydes and ketones [28]. In seafood, Feng et al. (2022) applied GC–MS to smoked perch, identifying 85 volatile compounds. Principal component analysis (PCA) and PLS-DA helped narrow down 25 key odorants, including hexanol (associated with fresh odor), aldehydes formed during drying, and phenols/furans enhanced by smoking [29]. GC–MS has also been applied in meat studies. Zhang et al. (2025) combined GC–MS with e-nose data to differentiate between fermented goat and lamb meat. The GC–MS analysis identified 18 signature volatiles, while e-nose readings achieved rapid and accurate meat-type classification [30]. Similarly, for golden pompano prepared using different cooking methods, GC–MS detected 108 aroma compounds and, through OPLS-DA, pinpointed 12 markers that explained the flavor differences among steaming, roasting, and air frying treatments [31]. Across these examples, GC–MS—especially when paired with multivariate statistical tools—has consistently demonstrated strong capabilities in distinguishing food samples based on volatile profiles. Its ability to link processing techniques to changes in aroma compounds makes it a key tool for flavor characterization in foods, particularly in fermented and aquatic products.

Gas chromatography-ion mobility spectrometry (GC–IMS)

GC–IMS combines two powerful techniques: traditional gas chromatography and ion mobility analysis. First, volatile compounds are separated by gas chromatography and then gently ionized, usually with a soft ionization source. These ions are pushed through a drift tube by a weak electric field, where they are sorted according to their size, shape, and charge. The result is a two-dimensional spectrum (retention time $\times$ drift time) that allows fast, sensitive, and high-throughput analysis of volatile compounds, even when present at very low levels. Because GC–IMS works under mild conditions and requires little sample preparation, it is especially well suited for quickly screening aroma profiles [32].

GC–IMS is especially useful for detecting small volatile molecules such as aldehydes, ketones, and alcohols. Although it provides less structural detail than GC–MS, it excels in speed and simplicity. For example, Wang et al. (2023) applied HS–GC–IMS to sun-dried citrus peel, detecting over 100 volatile peaks, 82 of which were identified—including various aldehydes, alcohols, terpenes, and ketones [33]. Miao et al. (2024) used a similar approach to study five types of fermented black beans, identifying 91 VOCs and using relative odor activity values (ROAVs) to highlight 23 key odorants [34]. Both studies showed strong agreement between GC–IMS and GC–MS results, with several compounds—such as non-

anal, hexanal, and eucalyptol-detected by both methods. In seafood, GC-IMS identified 72 volatile compounds in golden pompano, complementing the 108 identified by GC-MS [31]. The technique is especially valuable for real-time monitoring of VOC changes during fermentation or to distinguish geographical origins. Its colorful output maps can be analyzed using PCA or heatmaps to visualize sample differences. For example, GC-IMS profiles of citrus peels dried using different methods showed distinct patterns, with PCA results highlighting terpenes and furanones as key markers [35]. Overall, GC-IMS provides a rapid, high-throughput alternative or complement to GC-MS in flavor profiling.

Electronic sensing systems for flavor evaluation

Artificial sensory systems such as E-noses and E-tongues are designed to mimic human olfactory and gustatory perceptions and offer rapid, objective evaluations of food flavor. The E-nose consists of an array of gas sensors, typically metal oxide semiconductors or conductive polymers, that detect volatile compounds and generate digital signal patterns. The E-tongue uses electrochemical sensors, often potentiometric or voltametric electrodes, to respond to non-volatile tastants like salts, sugars, acids, and amino acids. Both tools generate multidimensional “flavor fingerprints” that can be analyzed using pattern recognition techniques such as PCA, LDA, PLS-DA, and neural networks. In flavoromics, E-sensing technologies are widely used for rapid screening and serve as complementary tools to traditional chromatography [36]. Although E-noses and E-tongues do not identify specific compounds, they are particularly effective for real time monitoring of freshness, spoilage, and quality changes due to their fast response and minimal sample preparation requirements.

Electronic nose (E-nose)

The E-nose mimics the human sense of smell using sensor arrays that respond to complex aroma profiles in the headspace of food samples. While it cannot identify individual compounds, it excels at rapidly capturing the overall volatile signature of a product. With advantages including high throughput, non-destructive analysis, and low cost, the E-nose is well suited for routine screening of aroma differences across large sample sets, particularly during food processing or spoilage [36]. For example, Zhang et al. (2025) demonstrated that an E-nose could effectively differentiate between fermented goat and lamb meats. PCA revealed distinct clustering patterns that clearly separated the two meat types, and these results were consistent with the key volatile compounds identified by GC-MS analysis [30]. In seafood research, Feng et al. (2022) used E-nose data to cluster golden pompano based on cooking methods. Boiled, steamed, and microwaved samples formed one group, while roasted and air-fried samples clustered separately. These findings corresponded with GC-MS results (highlighting differences in aldehydes and phenols) and sensory panel evaluations [29]. Chen et al. (2023) showed that GC-IMS and *uf*-GC E-nose could distinguish four Chinese freshwater fish species, both raw and cooked. PCA revealed clear clustering of VOC profiles, which aligned with key volatiles such as aldehydes, alcohols, ketones, and esters. These results highlight the potential of these techniques for flavor profiling and species classification [37]. When used alongside GC-MS, the E-nose helps associate specific sensor responses with chemical markers, forming a valuable bridge between instrumental detection and human perception.

Electronic tongue (E-tongue)

The E-tongue simulates human taste perception through sensor arrays that respond to primary taste modalities, including sweetness, sourness, bitterness, saltiness, umami, and astringency. It

typically employs electrochemical sensors, such as lipid/polymer membranes or metal electrodes, that generate electrical signals when exposed to specific tastants. Unlike chemical analysis, the E-tongue does not identify molecular structures, but it offers consistent and reproducible quantification of taste profiles [38].

E-tongues have been applied in a wide range of foods, from dairy and tea to fermented products and beverages, for tasks including taste profiling, quality control, and fermentation monitoring. For example, Chi et al. (2024) employed an E-tongue in combination with GC-MS, ion mobility spectrometry (IMS), and an E-nose to characterize regional variations in raw milk samples from different parts of China. PLS-DA of the E-tongue responses effectively differentiated samples by geographic origin, and the sensory profiles were further supported by key volatiles identified through chromatographic techniques. This multimodal strategy demonstrated the E-tongue's potential for authenticity verification in dairy products [39]. Similarly, Lv et al. (2025) employed E-tongue combined with untargeted metabolomics to study black mulberry juice fermented by single or mixed LAB. The results showed that mixed culture fermentation enhanced metabolite diversity and reduced bitterness. A decrease in astringency detected by the E-tongue correlated with lower levels of polyphenols and organic acids, as determined by LC-MS [40]. These findings illustrate how E-tongue data, when integrated with metabolomic insights, can effectively track sensory modulation during food fermentation processes. While less commonly used than E-noses, E-tongues are increasingly adopted in research on dairy, fermented foods, and functional beverages. Their ability to quantify liquid-phase taste components makes them a powerful complement to GC-MS/IMS in capturing the full sensory profile of food.

Multimodal data integration for flavor interpretation

Modern flavoromics increasingly integrates data from multiple sources, including GC-MS, GC-IMS, E-nose, and E-tongue, to achieve a more comprehensive understanding of flavor. This multimodal approach links molecular-level information from chromatographic platforms with perceptual-level responses from sensory devices, providing both chemical precision and holistic evaluation. In practice, integration is typically achieved through two strategies. At the feature level, descriptors from each modality are merged into a single feature matrix before model training, enabling algorithms to capture cross-modal interactions. At the decision level, independent models are developed for each modality, and their outputs are combined using ensemble methods such as weighted voting or stacking, which enhances robustness and mitigates noise from individual modalities. These integrative strategies are particularly effective in food quality control and flavor development, where connecting specific chemical signatures with overall sensory impressions is essential. Ensuring methodological transparency in applying such approaches is critical for reproducibility and aligns with best practices in data-driven food research [41].

Combining GC-MS and E-nose data balances chemical specificity with holistic aroma profiling. For example, in a study on Zhenjiang vinegar, Wang et al. (2022) identified 55 volatile compounds using GC-MS, of which 14, including furfural, acetic acid, and several pyrazines, were associated with product quality. In parallel, the E-nose effectively clustered samples by grade based on PCA results [42]. Fei et al. (2022) investigated the dynamic changes in volatile flavor compounds during the roasting of *Crataegi fructus* using an E-nose and HS-GC-MS. A total of 54 volatiles were identified, with aldehydes, ketones, esters, and furans showing significant variation. Grey relational analysis revealed strong correlations between 25 compounds and odor perception

Table 2
Representative machine learning models and their applications in food flavor prediction.

Model	Food Type	Model Task Type	Accuracy / R^2	Application scenarios	Advantages	Limitations	Reference
SVM	Beef	Classification	95.89 %	Widely applied to classification of food flavor profiles (meat, liquor) and regression modeling of fruit flavor compounds and sensory attributes	High accuracy with small/medium datasets; effective for nonlinear problems; good generalization	Requires careful parameter tuning; computationally expensive for large datasets; limited interpretability	Binson and Thomas (2023)
SVM	Chinese baijiu	Classification	97.7 %				Li et al. (2022)
SVM	Blueberry	Regression	$R^2 = 0.87$				Liu et al. (2023)
DT	Meat	Classification	89 %–94 %	Used for classification of volatile compounds and identification of key flavor contributors in meat and cereals	Simple and intuitive; strong interpretability; easy to implement	Prone to overfitting; lower accuracy compared with ensemble methods	Tuwani et al. (2019)
DT	Rice	Classification	90.8 %				Yang et al. (2024)
kNN	Chinese baijiu	Classification	98.38 %	Applied in similarity analysis and classification of fermented foods, alcoholic beverages, and condiments	Easy to understand; no training required; effective for small datasets	Performance depends heavily on dataset size and distance metric; poor scalability in high-dimensional data	Wu et al. (2019)
kNN	Beer	Classification	68.6 %				Wang et al. (2021)
KNN	Soybean paste	Classification	$R^2 = 0.93$	Applied in similarity analysis and classification of fermented foods, alcoholic beverages, and condiments	Strong capability in handling high-dimensional data; robust and resistant to overfitting	Less interpretable; relatively high computational demand	Jiang et al. (2025)
RF	Beer	Classification	94.4 %				Men et al. (2017)
RF	Chinese baijiu	Classification	100 %				Li et al. (2017)
RF	Yogurt	Regression	93.8 %	Applied to regression of fruit sensory traits, prediction of beer flavor scores, and classification of meat quality	High predictive accuracy; handles complex feature interactions; robust to missing data	Requires extensive hyperparameter tuning; limited interpretability	Tian et al. (2020)
RF	Chinese baijiu	Feature selection	$R^2 \geq 0.9$				Niu et al. (2025)
XGBoost	Blueberry and tomatoes	Regression	$R^2 = 0.87$				Colantonio et al. (2022)
XGBoost	Beer	Regression	$R^2 = 0.75$	Efficiently applied to classification of dairy hydrolysates, sweet vs. non-sweet molecules, and beverage or aroma categories	Fast training; scalable to large datasets; efficient for high-dimensional sparse data	May overfit on small datasets; black-box nature limits interpretability	Schreurs et al. (2024)
XGBoost	Meat	Classification	99.9 %				Son et al. (2024)
LightGBM	Milk protein hydrolysate	Classification	90.3 %				Yu et al. (2023)
LightGBM	Sweet compounds	Classification	96.1 %	Applied to flavor quality prediction and nonlinear modeling, including regression of fruit juice and tea sensory attributes, and classification of Maillard reaction products in meat	Captures complex nonlinear patterns; flexible; strong predictive performance	Requires large training datasets; training can be time-consuming; limited transparency	Lee et al. (2022)
LightGBM	Perfumes and liquors	Classification	94.8 %				Cao et al. (2022)
ANN	Fruit juice	Regression	$R^2 = 0.95$				Yang et al. (2023)
ANN	Beef tallow	Classification	>90 %	Effective for extracting features from high-dimensional structured data (spectra, GC–MS fingerprints), used in coffee, fish, and oil flavor classification	Learns hierarchical features automatically; strong performance with structural/visual data	High computational cost; requires large datasets; interpretability challenges	Cui et al. (2022)
ANN	Green tea	Regression	$R^2 = 0.875$				Yu et al. (2017)
CNN	Coffee	Classification	75–77 %				Chang et al. (2021)
CNN	Salmon	Classification	91.11 %				Sun et al. (2023)
CNN	Peanut oil	Classification	93 %				Bi et al. (2020)

($r > 0.9$), validating the combined use of E-nose and GC–MS for odor fingerprinting and quality control in thermal food processing [43]. GC–MS identifies the chemical basis of flavor, while the E-nose assesses its sensory impact, together providing a more comprehensive evaluation of food aroma. Although less commonly used, the combination of GC-IMS and the E-tongue presents promising potential for simultaneous analysis of both aroma and taste. For instance, Fu et al. (2024) evaluated deodorized rice field carp treated with perilla and cooking wine using GC-IMS, E-nose, and E-tongue. Such findings demonstrate how integrating volatile and taste data provides a clearer picture of consumer flavor perception [44]. While GC-IMS offers rapid profiling of volatiles, the E-tongue contextualizes taste response, enabling a more accurate prediction of flavor acceptability. The integration of these approaches has shown preliminary yet notable value in practical

market applications. Shen et al. (2023) characterized the flavor of five commercial grilled lamb shashliks using E-nose, E-tongue, GC-IMS and GC–MS [45]. Zhou et al. (2024) evaluated the flavor differences of yak jerky from three muscle sources using E-nose, E-tongue, GC–MS, and GC-IMS [46]. These findings underscore how multimodal data integration can significantly enhance market relevance and inform strategies for personalized food design.

In summary, analytical platforms such as GC–MS, GC-IMS, E-nose, and E-tongue generate heterogeneous data streams that vary in dimensionality, resolution, and noise characteristics. These properties strongly affect preprocessing requirements and influence the suitability of different ML models. A clear understanding of data acquisition methods therefore provides the foundation for selecting appropriate modeling strategies, which will be discussed in the next section.

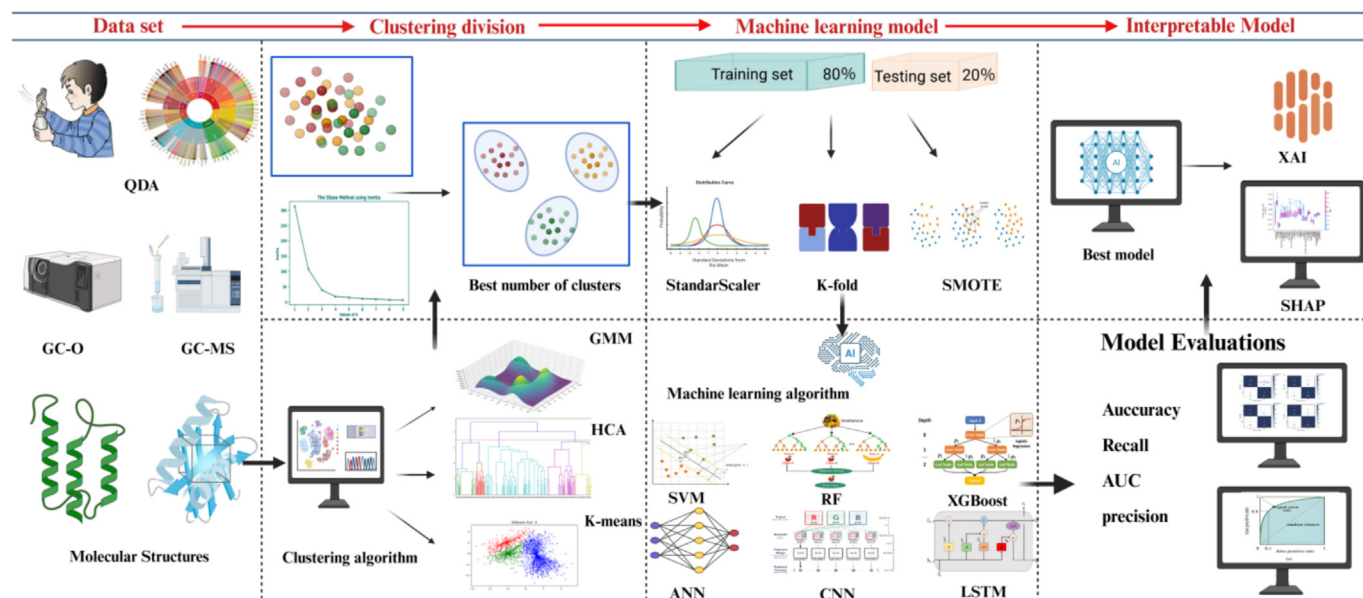


Fig. 1. Integrated machine learning workflow for food flavor prediction and model interpretation.

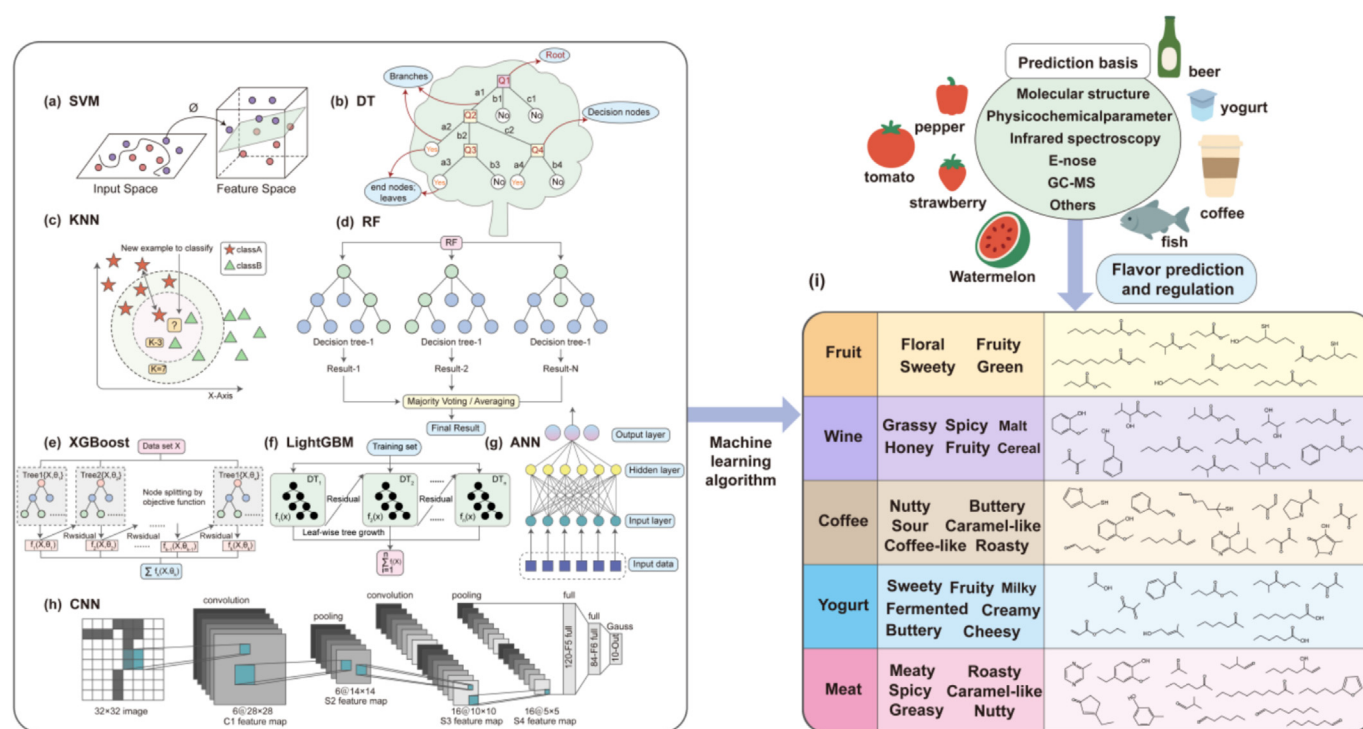


Fig. 2. Schematic overview of machine learning models and their applications in food flavor prediction and regulation. (a-h) Commonly used machine learning algorithms in food flavor research include support vector machine (SVM); decision tree (DT); k-nearest neighbors (KNN); random forest (RF); extreme gradient boosting (XGBoost); light gradient boosting (LightGBM); artificial neural networks (ANN) and convolutional neural networks (CNN). These models utilize a variety of input features, such as molecular structures, physicochemical parameters, spectral data, e-nose signals, and GC-MS profiles, to identify key flavor compounds and predict sensory attributes. (i) ML leverages flavor datasets collected from four key dimensions to predict and optimize the flavor profiles of foods.

Machine learning and explainable AI for food flavor modeling

Building on the data characteristics outlined above, this section introduces ML models and links their methodological strengths to the types of flavor data they are best suited to address. Traditional flavor assessment methods rely heavily on chemical analysis and human sensory evaluation, which, while essential, face limitations in handling complex, high-dimensional, and nonlinear flavor data.

To address these challenges, machine learning (ML) has emerged as a powerful set of tools capable of extracting patterns, modeling sensory responses, and enabling predictive control across food systems. This chapter presents an overview of key ML approaches used in flavor research, ranging from traditional supervised learning to advanced deep learning and explainable AI (XAI) techniques (Table 2). As shown in Fig. 1, ML techniques excel at modeling complex, high-dimensional datasets common in flavor research,

offering improved accuracy, real time processing, and the ability to adjust flavor profiles dynamically. In the final stage, explainable AI tools such as SHAP are used to visualize how key flavor compounds contribute to sensory attributes, enabling both accurate prediction and deeper understanding [47].

Traditional supervised learning methods

Support vector machines (SVM) in flavor prediction

SVMs are supervised learning models originally developed for binary classification and regression (Fig. 2a). They operate by mapping input data into a high-dimensional space, often through kernel functions such as the radial basis function (RBF), and then identifying the optimal separating hyperplane or regression boundary that maximizes the margin between classes. Because they rely on a limited set of support vectors, SVMs are particularly effective in handling small, high-dimensional, and nonlinear datasets [48].

SVMs have been widely used in flavor prediction tasks, including classifying products by taste profiles and predicting sensory scores. For example, Binson and Thomas (2023) applied an SVM model to classify beef samples as fresh or spoiled using data collected from a mobile E-nose system, achieving a classification accuracy of 95.89 % with 10-fold cross-validation [49]. Li et al. (2022) used SVM to classify the flavor types of Chinese Baijiu using comprehensive GC $\times$ GC-TOF-MS data, reaching an accuracy of 97.7 % [50]. In both cases, kernel types and model parameters were optimized to improve prediction performance. SVMs are also commonly used for regression tasks (SVR), especially in small datasets. A recent study on blueberries optimized an SVM model using particle swarm optimization (PSO-SVM) to predict sensory evaluation scores. This model achieved an RMSE of ~ 0.45 and an MAE of ~ 0.40 , significantly outperforming deep learning models like LSTM (RMSE = 1.22) and CNN (RMSE = 0.96) under the same conditions [51]. These findings underscore the strength of SVM in scenarios with limited data. Despite their strong performance, SVMs are often criticized for their lack of interpretability due to their high-dimensional decision boundaries. Their predictive accuracy depends heavily on proper feature scaling and careful kernel tuning [52]. In studies on the flavor classification of Baijiu, the SVM model has achieved reasonably good results. However, given the large scale of the data, SVM suffers from low training efficiency and high computational cost, making it less stable compared to ensemble learning methods [53]. While SVMs are considered “black-box” models compared to more interpretable algorithms like decision trees, they often deliver robust results, especially in nonlinear flavor classification and regression tasks involving limited or complex datasets. However, their inability to provide clear rules or feature importance limits their usefulness for mechanistic insights. In practice, SVMs often serve as reliable benchmarks, and while other models (like RF or deep learning) may outperform them with larger datasets, SVMs remain a valuable tool in flavor research.

Decision trees (DT) for taste classification

DTs classify or predict outcomes by recursively splitting data based on feature thresholds, resulting in a tree, like structure of simple “if-then” rules (Fig. 2b). Each node evaluates a specific variable, such as the concentration of a volatile compound or a sensor reading, and directs the sample toward a prediction at a terminal leaf [54]. This structure makes DTs highly interpretable, as domain experts can easily trace which features contribute to each decision.

Decision trees do not require complex optimization, making them versatile for handling mixed data types and identifying key predictors. However, single trees tend to overfit and generally underperform compared to ensemble models. In flavor science, DTs are used both independently and as base learners in ensemble methods. For example, Tuwani et al. (2019) used a DT to classify

molecules as “bitter” or “sweet” based on molecular descriptors. The model provided an intuitive mapping from structure to taste [55]. Similarly, Yang et al. (2024) used Tree Bagger to model rice taste based on physicochemical data, achieving 90.8 % accuracy, slightly below SVM but above other models. Despite lower accuracy, decision trees offer strong interpretability [56]. A recent study on Baijiu used an AdaBoost based DT model to identify key flavor compounds [53]. In general, DTs offer a clear trade off: while they may fall short of ensemble or kernel methods in accuracy, they are easier to interpret and can highlight influential features, which is valuable in flavor research and product development.

k-Nearest neighbors (kNN) for flavor similarity modeling

kNN is a non-parametric, instance-based algorithm that classifies or predicts a sample by identifying the k nearest neighbors in the training set, typically using Euclidean distance. It assigns the most frequent label in classification tasks or the average value in regression tasks [16]. The method is conceptually simple and does not require an explicit training phase, but it is sensitive to feature scaling and irrelevant variables (Fig. 2c).

kNN is often used as a baseline in flavor research, or combined with dimensionality reduction techniques for sensor or chemical data. For example, Wu et al. (2019) applied kNN to classify the aroma types of Chinese Baijiu using E-nose data. After preprocessing the signals with fuzzy discriminant PCA, their model achieved 98.38 % accuracy on the test set [57]. Wang et al. (2021) used kNN to classify 301 beer flavor compounds using molecular descriptors. Compared to RF (accuracy 68.6 %, $R^2 = 0.96$), kNN performed worse, highlighting its limitations in more complex scenarios [58]. In a study evaluating the quality of commercial soybean paste, Jiang et al. (2025) implemented a k -nearest neighbor (kNN) regression model using multi-sensor data. While the kNN model achieved a high R^2 of 0.9999 on the training set, its performance deteriorated on the test set ($R^2 = 0.9308$; RMSE = 6.53), indicating overfitting. The authors noted that due to its reliance on local similarity, kNN may struggle to capture global nonlinear patterns in complex food systems, making it less effective than models like SVR in predicting comprehensive flavor quality [59]. kNN generally yields moderate accuracy in flavor prediction tasks, often between 80 and 90 percent. It performs best with large, well labeled datasets where local similarity matters. Although it lacks interpretability and is sensitive to feature selection, its simplicity makes it a useful baseline. Poor performance with kNN often indicates the need for more advanced models such as SVM or random forest.

High-performance models for multi-dimensional flavor prediction

Random forest (RF) for flavor feature selection

RF are ensemble learning models that combine multiple decision trees to improve performance and stability. Each tree is trained on a different subset of data and features using bagging and random sampling techniques (Fig. 2d). Predictions are made by majority vote (classification) or averaging (regression). RF models balance the interpretability of decision trees with improved accuracy and reduced overfitting [48].

RFs are especially useful in flavor research due to their ability to handle high-dimensional data and multiclass problems. They also offer feature importance rankings, making them both predictive and somewhat interpretable. For example, Men et al. (2017) used RF to classify beer samples using data from combined E-nose and tongue sensors. Although SVM and ELM achieved slightly higher accuracy (96–98 %), RF still performed well at 94.4 % and was noted for its robustness [60]. Li et al. (2017) employed a QCM-based E-nose in combination with several classifiers, including RF, SVM, and discriminant function analysis (DFA). Among all tested models, RF achieved the highest classification accuracy of 100 %, SVM with

sigmoid kernel (96.1 %), SVM with linear kernel (89.5 %), and DFA (87.6 %). These results underscore the robustness and predictive superiority of RF models in distinguishing subtle flavor differences across complex aroma profiles [61]. Tian et al. (2020) applied RF regression to predict yogurt flavor acceptability using E-nose data and achieved 93.8 % accuracy, surpassing simpler models. In addition to prediction, RF is often used for variable selection [62]. For instance, Niu et al. (2025) applied an RF model with 500 trees to GC–MS data from Chinese Baijiu and successfully identified 19 key volatiles that distinguished between basic and derived flavor types. Overall, RF models consistently deliver high predictive power ($R^2 \geq 0.9$ and accuracy above 90 %) in flavor studies [63]. While less interpretable than single decision trees, they remain one of the most effective and reliable tools for both classification and feature analysis in complex food systems.

Extreme gradient boosting (XGBoost) for sensory prediction

XGBoost is a tree-based ensemble method that builds additive models by sequentially minimizing the residuals of prior predictions. Based on the gradient boosting framework, it incorporates regularization, efficient split-finding, and parallel computation for improved accuracy and scalability (Fig. 2e). These features contribute to its exceptional performance on structured data, particularly for tasks involving high dimensional, nonlinear, or incomplete inputs. However, the model is prone to overfitting on small datasets and requires careful tuning of hyperparameters.

In flavor research, XGBoost has become a widely used tool for modeling relationships between chemical composition and sensory perception. For example, Colantonio et al. (2022) evaluated 18 machine learning models, including XGBoost, using targeted metabolomic profiles of tomatoes and blueberries to predict sensory attributes such as sweetness, sourness, umami, and overall liking. XGBoost achieved an R^2 of 0.87 for predicting blueberry sourness, significantly outperforming simpler linear models, which yielded an R^2 of 0.46 [64]. Similarly, Schreurs et al. (2024) trained XGBoost regressors on chemical data from 250 commercial beers to predict 26 sensory descriptors. Compared with linear models, tree-based algorithms, particularly gradient boosting, showed superior performance, with the best model reaching a validation R^2 of 0.75 [14]. Although XGBoost was slightly outperformed by a finely tuned gradient boosting regressor, it still surpassed traditional statistical methods, confirming its effectiveness in modeling complex chemical-sensory relationships. In classification tasks, XGBoost has also demonstrated strong performance. In a recent study on emulsified pork sausages, Son et al. (2024) combined hyperspectral imaging (1000–2500 nm) with XGBoost to predict residual nitrite content non-destructively. The XGBoost model achieved an R^2 of 0.999 and an RMSE of 0.095, indicating near perfect predictive accuracy. This approach provides a rapid and real time tool for industrial meat quality monitoring, highlighting the potential of machine learning based hyperspectral analysis in food safety assurance [65]. Across both regression and classification tasks, XGBoost has proven to be a high-performing and versatile model in flavor prediction, typically reaching R^2 values ≥ 0.75 or classification accuracies in the 80–90 % range. Its ability to model nonlinear and high-dimensional data makes it particularly valuable in linking chemical signatures to sensory outcomes. However, researchers often combine it with feature selection methods or use model explanation tools to address its interpretability limitations and further improve performance and SHAP analysis helped identify important structural features associated with each taste profile [53].

Light gradient boosting machine (LightGBM) for efficient flavor modeling

LightGBM is an efficient gradient boosting framework that constructs decision trees using a leaf-wise growth strategy combined

with histogram-based feature binning (Fig. 2f). It is optimized for speed and memory usage, making it particularly suitable for large-scale, high-dimensional datasets. Like XGBoost, LightGBM handles missing values and categorical features natively and delivers comparable accuracy. However, its aggressive leaf-wise tree growth can lead to overfitting in smaller datasets if regularization is not properly applied [66].

In flavor prediction, LightGBM has demonstrated strong results in both classification and regression. Yu et al. (2023) developed a LightGBM classifier to identify bitter peptides in milk protein hydrolysates, using features such as amino acid composition, dipeptide frequency, and physicochemical properties. The model reached 90.3 % accuracy and an AUC of 0.891, demonstrating its effectiveness in peptide level taste prediction. Important features included the proportion of hydrophobic amino acids, a known bitterness marker, demonstrating the model's ability to learn complex sequence-taste relationships from high-dimensional biological data [67]. Lee et al. (2022) integrated LightGBM into an ensemble model, BoostSweet, to predict compound sweetness using molecular fingerprints and alvaDesc descriptors. The LightGBM component achieved an AUROC of 0.961. Feature analysis identified hydrogen bond donor nitrogen atoms as key contributors to sweetness perception, providing interpretable insights for sweetener design. LightGBM has also shown strong performance in sensor-based flavor classification [68]. Cao et al. (2022) applied it to E-nose data to classify odor fingerprints in perfumes and alcoholic beverages, achieving an accuracy of 95.5 %, slightly higher than XGBoost (94.8 %) [69]. All models in the study exceeded 90 % accuracy, confirming the strength of gradient boosting methods in handling complex aroma data. These examples show that LightGBM is a reliable and efficient option for modeling flavor perception. When properly tuned, LightGBM and XGBoost often achieve comparable performance, with validation accuracies above 90 %. Due to its superior computational efficiency, LightGBM is especially attractive for large scale metabolomic or spectroscopic datasets. However, like XGBoost, it lacks inherent transparency. Therefore, SHAP and similar tools are often used to interpret the model's outputs and understand which features drive sensory predictions.

In summary, both XGBoost and LightGBM effectively model nonlinear relationships through ensemble learning and consistently outperform simpler methods in linking chemical composition with sensory attributes. As gradient boosting algorithms, they are now widely adopted in flavor research, enabling accurate prediction of taste and aroma from complex datasets, as demonstrated in recent studies [67–69].

Deep learning approaches

Artificial neural networks (ANN) for nonlinear flavor regression

ANN are computational models inspired by the structure of biological neural systems, composed of interconnected nodes (neurons) organized in layers that process information through weighted connections. ANN, especially multilayer perceptrons trained by backpropagation, are well suited for capturing the complex nonlinear relationships inherent in food flavor systems. By learning complex mappings from inputs such as chemical profiles or sensor signals to sensory outputs, ANNs can reveal interactions among aroma compounds or sensor features that simpler models often miss [48]. In flavor research, Yang et al. (2023) used a feed-forward ANN to combine data from E-nose and tongue sensors, aiming to predict consumer preferences for fruit juices. Their model achieved an impressive fit to hedonic scores ($R^2 \approx 0.95$, RMSE ≈ 0.04), effectively simulating human sensory perception from multi-instrumental signals [70]. Similarly, Cui et al. (2022) used an ANN to analyze GC $\times$ GC–MS data of Maillard-derived volatiles in beef tallow. With a multi-input, single output architecture,

the model achieved over 90 % accuracy in predicting unseen sensory samples. ANNs are versatile, supporting both regression tasks such as predicting liking scores and classification tasks such as grouping by flavor notes [71]. In the beverage sector, Yu et al. (2017) developed a hybrid PLS-ANN model to predict consumer acceptance of ready-to-drink green tea. The ANN enhanced model outperformed conventional PLS regression, with an R^2 of 0.875 compared to 0.733 and a lower RMSE of 2.41 % versus 3.53 %, demonstrating the advantage of capturing nonlinear patterns in flavor data [72].

Despite their strengths, ANNs also face several notable limitations. They typically demand large datasets for effective training, which are not always available in flavor research. When datasets are small or noisy, overfitting may occur, resulting in poor generalization. Moreover, the “black-box” nature of ANNs makes it challenging to interpret how input features drive predictions—a critical drawback in mechanistic flavor studies. Their high computational cost and the need for extensive parameter tuning further limit their practical utility. In our view, although ANNs are powerful tools, their application in flavor prediction should be complemented with explainable AI methods and supported by feature engineering strategies to enhance both interpretability and robustness.

Convolutional neural networks (CNNs) for spectral feature extraction

CNNs build upon traditional neural networks by leveraging the spatial or spectral structure inherent in flavor related data. Their convolutional and pooling layers allow them to detect local features such as overlapping spectral peaks or aroma, related texture patterns, making them especially effective for analyzing spectroscopic data, sensor heatmaps, or even food imagery [16].

In flavor modeling, CNNs have shown strong promise. For example, Chang et al. (2021) trained a deep CNN on near-infrared (NIR) spectra from ground coffee beans to classify seven distinct flavor profiles. Despite a relatively small dataset of 266 sample, their model achieved classification accuracies of 75–77 % and recall scores of 70–73 %, slightly outperforming SVM bench-

marks [73]. Sun et al. (2023) applied a one-dimensional CNN to hyperspectral reflectance data from salmon muscle to detect off flavors such as geosmin and 2-MIB. The model achieved a classification accuracy of 91.11 %, substantially higher than traditional methods including PLS-DA and SVM [74]. In another study, Bi et al. (2020) transformed GC–MS aroma profiles of peanut oil into fingerprint images, then used a CNN to classify aroma types with approximately 93 % accuracy, far surpassing simpler models [75]. These results demonstrate CNNs’ ability to identify meaningful patterns even in limited data scenarios. Their performance improves significantly with larger datasets.

In summary, CNNs are particularly adept at extracting subtle spectral or spatial patterns tied to flavor perception. Whether applied to beverages like coffee or foods like fish and oils, CNN based models typically achieve accuracies between 75 % and 93 %, showcasing their practical utility for advanced flavor prediction.

Explainable AI (XAI) in flavor prediction

Despite the remarkable predictive power of ensemble and deep learning models, their “black-box” nature remains a critical limitation for flavor science. In practical applications, stakeholders, including product developers, sensory scientists, and regulatory authorities require transparency regarding how predictions are made, especially when flavor-modulating decisions influence product labeling, consumer perception, or formulation safety. To address this, explainable artificial intelligence (XAI) tools have been adopted to extract interpretable insights from complex models [76]. Among these, SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) are the most widely used in food flavor studies. SHAP quantifies the contribution of each feature to a model’s output using game-theoretic principles, offering both global and local interpretability. LIME, on the other hand, builds simple surrogate models near a prediction to highlight locally important features, making it suitable for visualizing spectral or image-based predictions [77].

Table 3

Summary of machine learning approaches for predicting food flavor attributes from diverse data sources.

Prediction Basis	Utilization Purpose(s)	ML	Reference
Molecular Structures	Bitterness prediction from molecular	AdaBoost + ECFP (QSAR)	Iwata. (2025)
	Bitterness and sweetness classification; feature selection	Ridge Logistic Regression + Chemopy; Random Forest; AdaBoost	Tuwani et al. (2019)
	Bitterness/sweetness classification	Ensemble (SVM + RF + DNN)	Tuwani et al. (2019)
	Sweet/bitter classification with interpretability	XGBoost + SHAP	Ferri et al. (2024)
Physicochemical Properties	Multi-label prediction of flavor notes (e.g., fruity, meaty, roasted)	Random Forest + kNN + RDKit + ECFP	Herrera-Rocha et al. (2024)
	Aroma intensity prediction based on 3D molecular structure	POI-3DGCN	Liu et al. (2022)
	Aroma persistence	Descriptor-based ML	Qiu et al. (2022)
	Prediction of odor descriptors from structural and electrostatic features	CNN	Zhang et al. (2023)
	Interpret model decisions	XGBoost + SHAP	Maroni et al. (2022)
	Prediction of aroma partition coefficients in dairy matrices	Random Forest, XGBoost + Molecular Descriptors (logP, MW, density)	Anker et al. (2024)
	Prediction of volatile compound distribution using thermodynamic and composition data	ML + Multivariate Statistics (with pH, moisture, fat, protein, sugar)	Goel et al. (2023)
	Prediction of red wine sensory characteristics from physicochemical indicators	1D CNN vs. SVM, Random Forest	Mittal et al. (2024)
	Flavor optimization in beer based on volatiles and physicochemical data	ML model correlating 200 + volatiles, pH, sugar, ethanol with consumer scores	Schreurs et al. (2024)
	Classify pleasant vs. unpleasant odors using E-nose data	CNN	Wu et al. (2019)
Instrumental Data	Predict volatile concentrations and aroma intensity in beer from E-nose	ANN	Gonzalez Viejo & Fuentes (2020)
	Classify peanut oil flavor/quality from GC–MS fingerprints	CNN	Bi et al. (2019)
	Classify baijiu by flavor style and origin using GC × GC-TOF/MS	SVM	Li et al. (2022)
	Wine quality classification based on GC-IMS	ANN	Zhu et al. (2021)
	Beer type classification using fused E-nose and E-tongue data	SVM	Men et al. (2017)

Recent studies have demonstrated the utility of XAI in flavor research. For example, Pallante et al. (2024) applied SHAP to a multi-target classifier predicting bitterness, sourness, and umami from molecular descriptors. The SHAP analysis identified key descriptors such as ATSC0c and AATSC2are as critical for distinguishing between taste modalities, thus connecting molecular structure with perceptual outcomes [78]. Similarly, Li et al. (2025) employed SHAP to interpret a Baijiu flavor quality model, revealing that tetramethylpyrazine and diethyl succinate positively contributed to sensory quality, while ethyl linoleate had a negative effect. These insights helped validate expert knowledge and offered direction for flavor enhancement [53]. Arrighi et al. (2025) used LIME to interpret CNN-based predictions from hyperspectral data in quality control tasks. The visualization of high-impact image regions facilitated operator trust and industrial deployment [79].

In recent years, explainable artificial intelligence (XAI) methods such as SHAP and LIME have been increasingly applied in flavor science. SHAP has been used to identify structural features and molecular descriptors most predictive of sweetness and bitterness, while LIME has been applied to interpret models linking instrumental data (GC-MS, E-nose) with sensory attributes. Despite their value, both approaches have limitations. SHAP offers theoretically consistent feature attributions but is computationally intensive for large datasets, and its explanations can be overly complex for practical use. LIME, though more efficient and intuitive, may yield unstable results across repeated runs and is restricted to local approximations that do not always reflect the global behavior of the model. Within flavor research, SHAP is particularly suited to mechanistic studies requiring rigorous feature attribution, whereas LIME is more useful for rapid, case-specific interpretation of sensory or metabolomic models. Collectively, these methods help bridge the “black-box” gap in ML by improving transparency and aligning model outputs with domain knowledge in food flavor science. As flavor science evolves toward structure-based flavor design, the role of interpretability will become increasingly central to both predictive accuracy and scientific understanding.

Multidimensional data-driven flavor prediction via machine learning

Food flavor prediction using ML involves accurately capturing complex relationships between diverse datasets and sensory attributes [48]. This section provides an in-depth overview of recent progress, categorizing prediction strategies based on input data types, including molecular structures, physicochemical properties, sensory evaluation data, instrumental analyses, and multimodal data integration (Table 3). As shown in Fig. 2i, ML leverages flavor datasets collected from four key dimensions to predict and optimize the flavor profiles of foods.

Flavor prediction from molecular structures

Molecular structure is fundamental to flavor perception, shaping volatility, solubility, receptor interactions, and metabolic stability. Structural characteristics such as molecular geometry, side-chain substituents, aromatic rings, and conformational flexibility determine sensory outcomes including aroma intensity, sweetness, bitterness, and flavor persistence. These properties provide the foundation for machine learning approaches, which employ molecular descriptors and fingerprints to predict flavor attributes [7,48].

Early work in computational flavor science employed QSAR models and molecular fingerprints to link structure with taste. For instance, Iwata (2025) developed BitterPredict, an AdaBoost-

based framework using extended connectivity fingerprints (ECFP) that demonstrated the predictive utility of structural features for bitterness [80]. Tuwani et al. (2019) compiled a large dataset of bitter and sweet compounds, showing that combining 2D and 3D descriptors with Ridge Logistic Regression yielded strong predictive performance ($F1 = 0.816$). Boruta feature selection further identified structural motifs most relevant to taste [55]. Ensemble approaches such as e-Bitter further enhanced robustness by integrating multiple classifiers, achieving 93 % accuracy on BitterDB benchmarks [81].

More recent studies have shifted toward interpretability and multidimensional prediction. Ferri et al. (2024) applied XGBoost with SHAP analysis to associate bitterness and sweetness with specific substructures and functional groups, offering interpretable insights into molecular drivers of taste. Building on this, Herrera-Rocha et al. (2024) introduced FlavorMiner, a multi-label framework that predicted multiple flavor attributes (fruity, meaty, roasted) with an AUC of 0.88, illustrating the scalability of structure-based modeling [7].

Advances in deep learning and graph-based methods have further emphasized the role of structural information. Liu et al. (2022) proposed POI-3DGCN, which employed 3D molecular graphs to predict aroma intensity and revealed links between hydrophobic surface area, aromatic rings, and odor strength [82]. Qiu et al. (2022) introduced Odor-GCN, a graph neural network that directly predicted perceptual attributes such as pleasantness and familiarity from molecular graphs [83]. Extending this line of research, Zhang et al. (2023) developed Mol-PECO, integrating Coulomb matrices with Laplacian eigenvalue encodings to capture electrostatic and spatial features. This approach achieved an AUROC of 0.813, outperforming traditional 2D fingerprint-based classifiers [84].

In summary, molecular structure provides a mechanistic basis for accurate flavor prediction. Side-chain substituents modulate volatility and receptor binding, aromatic rings are associated with characteristic odor qualities, and flexible aliphatic structures contribute to both complexity and richness of sensory profiles. Incorporating these structural determinants into molecular descriptors has improved the accuracy and interpretability of ML models. Looking forward, progress will depend on combining advanced 3D descriptors with explainable AI to develop models that are not only predictive but also mechanistically informative.

Flavor prediction from physicochemical properties

Flavor is deeply influenced by the physical and chemical properties of food, such as pH, viscosity, water activity, sugar content, and volatile compound concentration. These factors affect how flavor molecules interact, how quickly they're released, and how they're ultimately perceived by the senses. For example, a thicker texture with higher viscosity can trap aroma compounds and reduce their release, while changes in pH can affect the ionization and volatility of key aromatic molecules. At the same time, sensory feedback from trained panels or consumers offers a subjective complement to data obtained from instruments. By integrating both types of data, machine learning models can link objective physical and chemical parameters with subjective flavor perception, enabling the prediction and refinement of specific flavor characteristics [16].

A representative study by Anker et al. (2024) demonstrated the feasibility of using regression-based machine learning models, such as random forest and XGBoost, to predict aroma partition coefficients in dairy matrices. The models incorporated molecular descriptors such as log P, molecular weight, and density, which reflect volatility and the potential for interaction with the matrix, and achieved high predictive accuracy with an R^2 of 0.99. In addition,

tion, explainable artificial intelligence methods confirmed these features as major contributors to the dynamics of flavor release. Building on this work, an earlier study by the same group expanded the set of input variables by integrating pH, moisture content, and macronutrient composition (fat, protein, and sugars) to predict the distribution of volatile compounds in diverse food systems. By combining thermodynamic parameters with machine learning algorithms and multivariate statistical techniques, the authors effectively captured the complex interactions between environmental factors and flavor retention [85]. In a related effort, Mittal et al. (2024) applied a one-dimensional convolutional neural network to predict the sensory characteristics of red wine based on physicochemical indicators such as pH, titratable acidity, residual sugar, and sulfur dioxide content. Their deep learning model outperformed conventional approaches, including support vector machines and random forest, suggesting that nonlinear relationships among input variables substantially influence flavor perception [86]. At the product development level, Schreurs et al. (2024) used machine learning to optimize beer flavor by correlating over 200 volatile compounds, together with pH, sugar content, and ethanol concentration, with consumer preference scores. The model not only identified the key drivers of positive flavor perception but also informed formulation strategies that were subsequently validated through experimental trials [14]. These findings underscore the practical potential of combining physicochemical data with machine learning to support targeted and data driven flavor optimization.

Together, these studies underscore a converging research paradigm wherein quantitative physicochemical features serve as foundational predictors for both flavor release mechanisms and sensory perception outcomes. By embedding these features into machine learning pipelines, researchers can model complex interactions between food and flavor with greater precision, transparency, and relevance to industrial applications.

Flavor prediction from instrumental data

Modern flavor analysis is increasingly powered by sophisticated instruments like E-nose, E-tongue, GC-MS, and GC-IMS. These technologies generate complex and high dimensional chemical data, which makes them well suited for machine learning applications. By objectively capturing volatile and nonvolatile compounds associated with aroma and taste, these tools generate rich datasets of measurable inputs that machine learning algorithms can learn to associate with sensory qualities [48].

Take the E-nose as an example, which is an array of gas sensors designed to mimic human olfaction. When paired with deep learning, it becomes a powerful tool for classifying aroma. Wu et al. (2019) used a CNN trained on E-nose data to distinguish between pleasant and unpleasant smells, achieving a remarkable 99.9 % accuracy. The results aligned with human perception and demonstrated that CNNs can identify subtle features that shape the way aromas are experienced [57]. Other studies have gone further, using low-cost sensor systems to predict flavor profiles. Gonzalez et al. (2020) developed ANN models using E-nose signals to estimate both the concentration of 17 volatile compounds and the sensory intensity of 10 aroma attributes in beer. The models closely matched lab and panel results, with correlation coefficients of 0.97 and 0.93, respectively [87]. These findings show that E-nose data, when interpreted through neural networks, can serve as a practical substitute for both analytical chemistry and human sensory panels. GC-MS remains the gold standard for volatile compound analysis and is now frequently paired with machine learning models in flavor research. Unlike targeted approaches, GC-MS produces full chromatograms, which are complex outputs representing a broad spectrum of compounds. These require pat-

tern recognition rather than simple quantification. In this space, deep learning excels. Bi et al. (2020) transformed GC-MS data from peanut oil into image like fingerprints and applied CNNs to classify samples by flavor and quality. The model achieved 93 percent accuracy, significantly outperforming traditional PLS regression, which often fails to capture the nonlinear and interdependent signals present in chromatographic data [75]. Similarly, Li et al. (2022) used GC $\times$ GC-TOF/MS data to classify Chinese baijiu samples by flavor style and geographic origin. Among the classifiers tested, support vector machines performed best, achieving an accuracy of 97.67 percent, which exceeded the 91.9 percent accuracy of random forests. GC-IMS, a faster and highly sensitive alternative to GC-MS, is gaining traction in flavor science. Its outputs are now regularly used in ML pipelines for product quality prediction [50]. For example, Zhu et al. (2021) compared six algorithms, including PCA-LDA, PLS-DA, kNN, SVM, XGBoost, and ANN, to classify wine into expert defined quality levels based on GC-IMS data. The ANN model came out on top, with a 95.4 % classification accuracy. This result underscores how more complex, nonlinear models can better capture subtle spectral patterns linked to sensory characteristics. When analyzing taste and aroma together, combining sensor data from multiple instruments can provide better predictions than relying on one alone [88]. Men et al. (2017) demonstrated this by combining E-nose and E-tongue data to classify five types of commercial beer. Traditional classifiers like SVM and random forest saw accuracy improve from about 85 % to 89 % when the data were fused, showing the advantage of capturing complementary taste and aroma information [60].

In summary, recent research demonstrates that machine learning techniques are highly effective in identifying flavor relevant patterns from instrumental data. These models can capture complex relationships between chemical signatures and sensory outcomes, enabling faster and more objective flavor assessment. With continued progress in deep learning, sensor integration, and feature selection, machine learning based flavor recognition is positioned to become a central tool in modern food science. From real time quality control in production to the design of consumer-oriented flavor profiles, the range of applications is both broad and rapidly expanding, particularly in scalable solutions for food quality monitoring.

Multi-source data fusion in flavor prediction

Flavor perception is a complex trait shaped by molecular structures, physicochemical properties, instrumental outputs, and human sensory evaluations. Each of these data dimensions captures a different aspect of flavor, yet they vary in scale, feature distribution, and noise. Molecular descriptors are often high-dimensional and sparse, instrumental platforms such as GC-MS, GC-IMS, and E-nose generate large but noisy fingerprints, and sensory data are comparatively low-dimensional and subjective. Integrating these complementary yet heterogeneous datasets has therefore become a central challenge in flavor modeling, driving research on multi-source heterogeneous data fusion [89].

Two primary strategies are commonly employed. Feature-level fusion merges descriptors from multiple modalities into a unified feature space before modeling, enabling algorithms to capture cross-domain interactions. For example, Men et al. (2025) combined HS-GC-IMS and E-nose data for origin authentication of *Amomi fructus*, achieving 97.96 % accuracy, markedly higher than any single modality [90]. Likewise, Yu et al. (2024) integrated peptidomics-derived molecular features into a LightGBM-based CPM-BP model to predict bitter peptides in milk hydrolysates, reaching a classification accuracy of 90.3 % and experimentally validating several predicted bitter peptides [91]. These examples highlight the capacity of feature-level fusion to exploit comple-

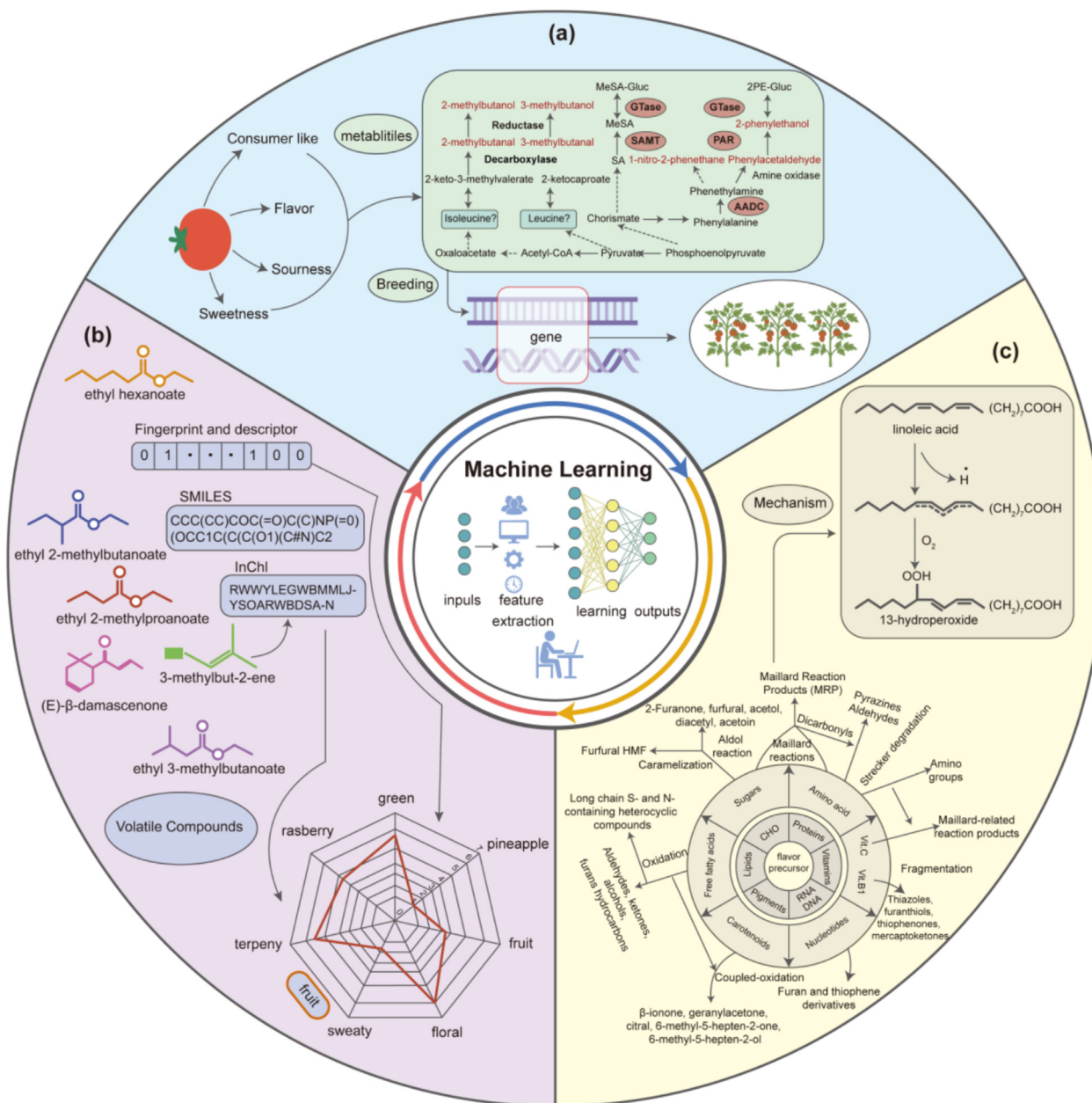


Fig. 3. Future perspectives in machine learning for food flavor research machine learning is expected to play an increasingly central role in advancing flavor science across multiple research frontiers: (a) Identification and Screening of Key Flavor Metabolites; (b) Genotype-Flavor Integration and Breeding Targets; (c) Flavor Compound Synthesis and Reaction Prediction.

mentary chemical and instrumental information, though the risk of dimensional imbalance and overfitting requires careful feature selection. Decision-level fusion represents an alternative strategy in which separate models are trained on individual modalities and their outputs are subsequently integrated through ensemble methods. This approach enhances robustness by reducing the influence of noisy features from any single data source. Schreurs et al. (2024) applied decision-level fusion by combining GC–MS volatile profiles with sensory evaluation data from 250 beer samples. Their ensemble model not only improved prediction accuracy but also identified trace volatiles with disproportionate sensory importance, demonstrating how fusion can enhance both robustness and interpretability [14].

Overall, multi-source heterogeneous data fusion provides a powerful framework to leverage the complementary strengths of

molecular, instrumental, and sensory datasets. Feature-level fusion captures fine-grained cross-modal interactions but risks dimensional imbalance, whereas decision-level fusion enhances robustness but may overlook subtle dependencies. Emerging hybrid approaches seek to combine these advantages, offering both accuracy and stability. As flavor science increasingly relies on multi-modal data, integrating heterogeneous sources will be essential to improving predictive accuracy, advancing mechanistic understanding, and supporting intelligent flavor design.

Further of machine learning in food flavor regulation

Flavor regulation, the purposeful control and modification of food flavor profiles, has emerged as a crucial focus within food science. Flavor plays a pivotal role in determining how well a food

product is received by consumers and ultimately shapes market trends. A pleasant and appealing flavor is often considered essential for any successful food item. Traditionally, food flavor has been regulated through methods such as controlling microbial activity during fermentation, adjusting processing techniques, or adding flavor enhancers. While these approaches can be effective, they are often time consuming, costly, and difficult to control with precision [64]. Recently, ML has gained traction as a powerful new tool for flavor optimization. Thanks to its strong computational capabilities, high efficiency, and greater controllability, ML offers a promising alternative to traditional methods and is attracting growing interest across the food industry. Recent developments in ML methodologies have significantly expanded the capacity for precise and efficient flavor regulation. Unlike flavor prediction, which seeks primarily to model and forecast sensory outcomes based on available data, flavor regulation aims actively to guide and manipulate flavor profiles through targeted interventions [92]. This section presents a detailed examination of current machine learning applications, focusing on metabolite screening, integration of genotype and flavor, prediction of chemical synthesis, and process optimization aimed at targeted flavor enhancement (Fig. 3).

Identification and screening of key flavor metabolites

ML has rapidly become a valuable tool in uncovering the chemical foundations of food flavor. Unlike traditional sensory panels, which are limited by scale and subjectivity, machine learning enables rapid and objective screening of flavor active metabolites at high throughput. Recent studies have demonstrated that ensemble models and neural networks can effectively analyze complex metabolomic data to identify the compounds most relevant to taste and aroma (Fig. 3a). Metabolomic datasets are typically high-dimensional, often containing thousands of detected features, many of which are highly collinear due to shared biosynthetic pathways. To address these challenges, dimensionality reduction and feature selection are commonly applied prior to model construction. PCA and PLS reduce dimensionality while preserving major sources of variance. Regularization techniques such as LASSO and elastic net penalize redundant variables and facilitate the identification of informative metabolites. Tree-based ensemble algorithms, including Random Forest and XGBoost, are particularly advantageous for metabolite screening because they inherently manage multicollinearity and provide measures of variable importance. Through these strategies, researchers can narrow analyses to subsets of metabolites with the greatest predictive value, thereby improving both interpretability and predictive performance [1].

In strawberries, researchers applied an XGBoost model to GC-MS volatile profiles from 148 cultivars and corresponding consumer sensory scores. Esters and lactones such as γ -dodecalactone and 5-methyl-5-hepten-2-one were identified as key predictors of fruity sweetness. Including these volatiles improved the explained variance in sweetness and liking by over 25 percent compared to using sugar and acid content alone, highlighting the role of minor compounds in shaping flavor perception [93]. Similarly, Ferré et al. (2022) applied a random forest model to link blueberry metabolomic profiles with sensory traits, identifying eight key terpenes, with linalool and d limonene playing major roles in floral and sweet notes. The model successfully distinguished high aroma genotypes and offered insights to support flavor focused breeding. Machine learning is also contributing to a deeper understanding of metabolite biosynthesis [94]. Liu et al. (2022) combined transcriptomic and metabolomic data from grape berries, using gradient boosting and network analysis to link terpene levels with gene expression. They identified six terpene synthase genes and three transcription factors involved in linalool

production, offering targets for flavor improvement through breeding [95]. Beyond single compounds, machine learning helps uncover metabolite clusters, or “flavor modules,” that jointly affect taste and aroma. In strawberries and tomatoes, network and sparse regression methods (WGCNA, LASSO) identified groups of esters, phenylpropanoids, and amino acid derivatives associated with high sensory scores. These models revealed key metabolic pathways, such as phenylalanine metabolism and branched chain amino acid metabolism, which influence flavor more effectively than traditional untargeted approaches.

In summary, machine learning driven metabolite screening represents a transition from merely measuring flavor to actively managing it. By accurately identifying the chemical components of taste and aroma and linking them to genetic, formulation, or processing factors, machine learning provides an unprecedented level of precision in shaping food flavor profiles.

Genotype-flavor integration and breeding targets

The integration of genomics and metabolomics through ML has opened new doors for mapping the genetic basis of flavor in crops, making it possible to breed for improved taste with greater precision. Flavor is a complex, polygenic trait that has historically been difficult to select for using traditional methods. By combining genetic markers with detailed metabolite profiles related to flavor, ML models can uncover the often-nonlinear relationships between a plant's genotype and its sensory characteristics. This allows breeders to predict flavor outcomes early in the breeding process, rather than relying on years of field trials and taste tests (Fig. 3b) [92].

To accelerate the identification of flavor enhancing genotypes, Colantonio et al. (2022) trained machine learning models on metabolic profiles, including volatiles, sugars, and acids, to directly predict consumer sensory ratings in tomatoes and blueberries. Their results showed that metabolic markers could explain 42–56 % of the variation in liking scores, significantly outperforming models based solely on genotype. This approach enables rapid screening of breeding lines for favorable flavor phenotypes, while maintaining a clear link to underlying genotypic variation through subsequent QTL and GWAS integration [64]. Fernie et al. (2022) used SHAP to interpret machine learning models in tomato and pepper, identifying genetic markers most influential on flavor compounds and thereby improving the decision-making process for breeders [96]. Fan et al. (2022) constructed a genotype-flavor interaction network in strawberries, linking genes, intermediates, and aroma compounds, and revealing new transcription factors involved in aroma biosynthesis. An emerging strategy is metabolomic selection, where ML models predict sensory scores directly from metabolite profiles [97]. Colantonio et al. (2022) demonstrated this in tomatoes and blueberries, with volatiles and sugars explaining 42–56 % of the variance in consumer liking. This approach provides a rapid chemistry-based alternative to traditional sensory panels and allows for tracing key metabolites back to their associated genetic loci. Once critical flavor metabolites are identified, linking them to genetic markers completes the pathway from phenotype to genotype, enabling targeted breeding strategies [64]. To develop cultivars that combine desirable flavor with agronomic performance, researchers are adopting multi-trait genomic prediction. Multitask learning models capture correlations among traits such as flavor, yield, and disease resistance, reducing trade-offs in selection. Additionally, graph-based methods like graph convolutional networks (GCNs) are being explored to map gene, metabolite, and flavor networks, revealing key regulatory nodes and complementing traditional QTL mapping. Importantly, once key metabolic predictors are identified, they can be traced back to their genetic origins via genomic association analyses, thereby closing the loop

from phenotype to genotype. This integration of metabolomic and genomic data offers breeders a data driven framework to select flavor rich cultivars with greater precision and efficiency.

Flavor compound synthesis and reaction prediction

Understanding and modeling the chemical reactions responsible for flavor formation is a core objective in food science. Essential processes such as the Maillard reaction, fermentation, and lipid oxidation produce both desirable aromas and undesirable off flavors. Accurate predictive models of these pathways allow for targeted control of processing conditions to enhance favorable flavors while suppressing unwanted compounds (Fig. 3c) [48].

ML has emerged as a powerful approach for modeling these multifactorial pathways, offering a faster and more scalable alternative to traditional trial and error methods. Recent studies have demonstrated that machine learning can not only predict how processing conditions influence flavor outcomes but also guide the design of new aroma molecules, marking the beginning of a data driven era in flavor chemistry. A central focus has been the Maillard reaction, which produces both savory and roasted notes in cooked foods. While traditional models like PLSR have linked process variables to flavor outcomes, they often require large datasets and struggle with complex, nonlinear interactions. ML, by contrast, can capture these dynamics more efficiently. For example, Cui et al. (2022) used an ANN to predict the sensory quality of beef Maillard reaction products derived from enzymatically treated tallow. Their model, trained on multiple reaction parameters, achieved over 90 % accuracy in forecasting sensory panel scores. This allowed them to identify an optimal combination of enzymes, papain and flavorzyme, which maximized desirable meaty notes while minimizing undesirable off flavors [71]. ML has also proven useful for improving food safety. Wu et al. (2023) used an ANN (BP-NN) model to predict acrylamide levels in fried dough by incorporating factors such as color, oil quality, and frying conditions. The model performed with a high R^2 of 0.972 and prediction errors of less than 0.01 %, providing a practical tool for reducing this potentially harmful compound while maintaining flavor quality [98]. Fermentation is another area where ML is making an impact. Du et al. (2025) applied GBDT to predict the formation of meat-like aromas during yeast fermentation. The model identified optimal pH, temperature, and aeration conditions for reproducing the aroma of canned meat, and it highlighted seven key volatile compounds, including pyrazines and furans, that contribute strongly to meat-like flavors. Notably, several of these compounds had received little attention in earlier studies [99]. Likewise, Schreurs et al. (2024) applied LightGBM to analyze GC-MS and sensory data from more than 250 beers. The model identified trace volatile compounds with disproportionately strong effects on consumer preference, and adjustments guided by these findings led to significant improvements in sensory ratings [14]. ML is also helping to reduce undesirable flavors. In seafood and algae-based foods, for example, minimizing fishy notes remains a major challenge. Sun et al. (2025) applied KNN to a large odorant database to predict fishy odor intensity from molecular structures. The model highlighted specific amines and aldehydes as major contributors to marine off-flavors [100]. By targeting these compounds, food engineers can redesign formulations or adjust processing conditions to achieve cleaner and more appealing flavor profiles. In fermented sauces and seafood products, similar ML approaches can guide modifications to fermentation time, microbial strains, or precursor concentrations to suppress undesirable odors such as sulfurous, ammonia-like, or rancid notes. Beyond predicting known reactions, ML is increasingly being applied to the design of entirely new flavor compounds. Advances in deep generative models now make it possible to create novel aroma molecules *in silico*. For example, Queiroz et al. (2023) developed an ANN trained on thousands of known fla-

vor compounds. The model captured structural features linked to specific sensory characteristics and generated new candidates consistent with desired flavor profiles. Interestingly, many of the compounds it identified were already in use in the food and fragrance industries, suggesting that the model effectively rediscovered fundamental chemical principles underlying desirable aromas [101]. Rodrigues et al. (2024) advanced this line of research by integrating ANN (GNN) with reaction optimization. Their system combined a molecule generator, a flavor intensity predictor, and a volatility model to iteratively design flavor compounds that aligned with target profiles [102].

Altogether, these advances are establishing a new paradigm in flavor science, where data-driven models not only predict which compounds will form under specific conditions but also guide interventions to improve taste, safety, and consumer acceptance. ML helps pinpoint critical control points by identifying the precursors, enzymes, or processing parameters that should be adjusted to optimize both flavor intensity and overall quality. Collectively, these strategies mark a transition from empirical flavor adjustment to data-driven control.

Conclusion and future perspective

This review highlights the increasing reliance of food flavor analysis on advanced sensory technologies, particularly GC-MS, GC-IMS, E-nose, and E-tongue. These techniques are often used in combination to capture comprehensive chemical and sensory profiles of foods. Nonetheless, several limitations remain: sensory methods retain elements of subjectivity, instrumental analyses generate high-dimensional data that are challenging to integrate, and the absence of standardized, large-scale datasets constrains reproducibility and generalizability. ML offers significant potential to address these challenges. Models such as SVM, DT, RF, LightGBM, XGBoost, ANN, and CNN have been successfully applied to predict and regulate flavor attributes across diverse food systems. ML has proven effective in identifying key volatile markers, linking microbial communities or crop genotypes to flavor phenotypes, and guiding the adjustment of processing parameters toward defined sensory outcomes. These data-driven approaches not only enable accurate characterization of flavor compounds but also provide mechanistic insights into metabolic and fermentation control, thereby helping to overcome the limitations of conventional methods. Looking forward, several avenues merit attention. The incorporation of XAI is expected to mitigate the “black-box” limitation of ML improving transparency and interpretability. The development of larger, standardized, and multi-modal flavor datasets will enhance model robustness and comparability. Advances in graph-based methods and 3D structure-aware learning will allow more precise modeling of molecular features critical to flavor perception. Finally, the application of ML to real-time flavor monitoring, personalized nutrition, and intelligent food design represents a promising frontier for both academic research and industrial practice. Within this trajectory, ML evolves from a passive predictive framework into an active tool for real-time flavor modulation and optimization.

Compliance with ethics requirements

This article does not contain any studies with human or animal subjects.

CRedit authorship contribution statement

Xinyu Ge: Methodology, Investigation, Visualization, Writing – original draft. **Yongjie Zhou:** Writing – review & editing. **Qing Li:**

Writing – review & editing. **Yuqing Tan:** Writing – review & editing. **Yongkang Luo:** Funding acquisition, Writing – review & editing. **Hui Hong:** Conceptualization, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Xinyu Ge is currently a first-year Ph.D. student in the Department of Food Science and Nutrition Engineering at China Agricultural University. During his master's studies, he published four first-quartile (Q1) SCI papers. His current research focuses on the flavor formation mechanisms and predictive modeling of Maillard reaction products derived from salmon by-products, using machine learning approaches.



Yongjie Zhou is currently a Ph.D. candidate in Food Science and Engineering at China Agricultural University. His research focuses on the development of animal-derived bioactive peptides, with particular emphasis on optimizing peptide preparation using machine learning approaches, improving peptide flavor through enzymatic techniques, and exploring the functional properties and underlying mechanisms of bioactive peptides.



Qing Li graduated from Nanjing University of Finance and Economics with a bachelor's degree and is currently pursuing a doctoral degree at China Agricultural University. My research focuses on the quality changes and early warning technologies of fine-cut bighead carp products during storage.



Yuqing Tan is an Associate Professor at the College of Food Science and Nutritional Engineering, China Agricultural University, and an IFT certified food scientist. She was selected for the 7th Youth Talent Support Program of the China Association for Science and Technology and serves as a post expert in sturgeon and salmon processing for the Beijing Fishery Innovation Team of the Modern Agricultural Industry Technology System. She received her M.S. and Ph.D. degrees from Mississippi State University and completed her post-doctoral research at the USDA Western Regional Research Center. Her research focuses on the functional properties of aquatic proteins, high-value utilization of aquatic by-products, and bioactivity evaluation of animal-derived peptides and natural compounds. Dr. Tan has published over 50 papers in international journals, led more than 15 national and provincial research projects, and received several awards for scientific and technological progress.



Yongkang Luo is a Professor at the College of Food Science and Nutritional Engineering, China Agricultural University, and Director of the National R&D Center for Freshwater Fish Processing Technology (Beijing). He has been recognized as one of Elsevier's Highly Cited Chinese Researchers in Food Science and Engineering for five consecutive years (2020–2024). He serves as the chief expert of the Fish Product Quality, Safety and Nutrition Communication Team of the Chinese Fisheries Society and as a member of the Sub-Technical Committee on Aquatic Product Processing of the National Technical Committee on Aquatic Product Standardization. His research focuses on aquatic product preservation, quality control, efficient resource utilization, and functional evaluation of animal protein and peptides. Dr. Luo has published over 500 papers, including more than 260 in international journals, and has supervised over 160 graduate students.



Hui Hong is a Professor at the College of Food Science and Nutritional Engineering, China Agricultural University. His research focuses on the nutritional quality, storage, preservation, and processing of aquatic products, as well as the high-value utilization of animal protein resources and bioactive peptides. He has published over 60 papers in leading international journals such as *Chemical Engineering Journal*, *Journal of Agricultural and Food Chemistry*, and *Food Chemistry*, with more than 8,000 citations (H-index 49). Dr. Hong serves as Guest Editor of *Food Chemistry* and on the editorial boards of several journals. He was listed among the World's Top 2% Scientists in 2023, 2024 and 2025 and has received multiple awards for scientific and technological progress.